Krishi Sahayak: AI-Powered Multilingual Farmer Advisory System

Comprehensive Technical Walkthrough, System Architecture & Operational Guide | **Version 2.0 (September 2026)** | Target: All 28 Indian States & 8 Union Territories

1. Executive Summary

1.1 Problem Statement

Indian agriculture sustains over **140 million farming households**, representing the backbone of national food security. Approximately **86% are smallholder and marginal farmers** operating on less than 2 hectares. Despite advancements in agronomic research, smallholder farmers face severe structural challenges:

- **Linguistic Exclusion & Illiteracy:** India has 22 official languages and hundreds of regional dialects. Most agricultural portals and mobile apps are authored in formal English or standardized Hindi, leaving regional language speakers and illiterate farmers unable to access verified guidance.
- **Pesticide Overuse & Pest Misdiagnosis:** Due to lack of real-time diagnostic support, farmers often consult commercial pesticide retailers, leading to chemical misapplication, hazardous over-spraying, environmental degradation, and debt cycles.
- **Severe Public Extension Deficit:** The ratio of Krishi Vigyan Kendra (KVK) extension officers to operational farmers is constrained (~1:1,200), causing critical advice delays during acute pest outbreaks or climate shocks.
- **LLM Hallucination Liabilities:** General-purpose AI models frequently hallucinate chemical dosages or invent plant treatments, posing direct threats to crop survival and farmer livelihoods.

1.2 Target User Base

- **Primary:** Smallholder and marginal farmers across all 28 Indian States and 8 Union Territories, spanning diverse literacy levels and regional languages.
- **Secondary:** Village Panchayat operators, Primary Agricultural Credit Society (PACS) workers, and KVK field extension staff assisting farmers in rural centers.

1.3 Core Value Proposition

Krishi Sahayak (कृषि सहायक) bridges this scientific extension gap with a tri-modal, 22-language native localized, RAG-grounded advisory platform:

🌱 Strictly Grounded RAG Intelligence

100% of advisory responses are retrieved from verified ICAR and State Agricultural University manuals via **ChromaDB**. Zero reliance on generative LLM training data.

📸 Two-Stage Visual Crop Diagnostics

GPT-4o Vision is strictly constrained to objectively describe visible plant symptoms. Those descriptions are then matched against ICAR manuals to formulate treatment protocols.

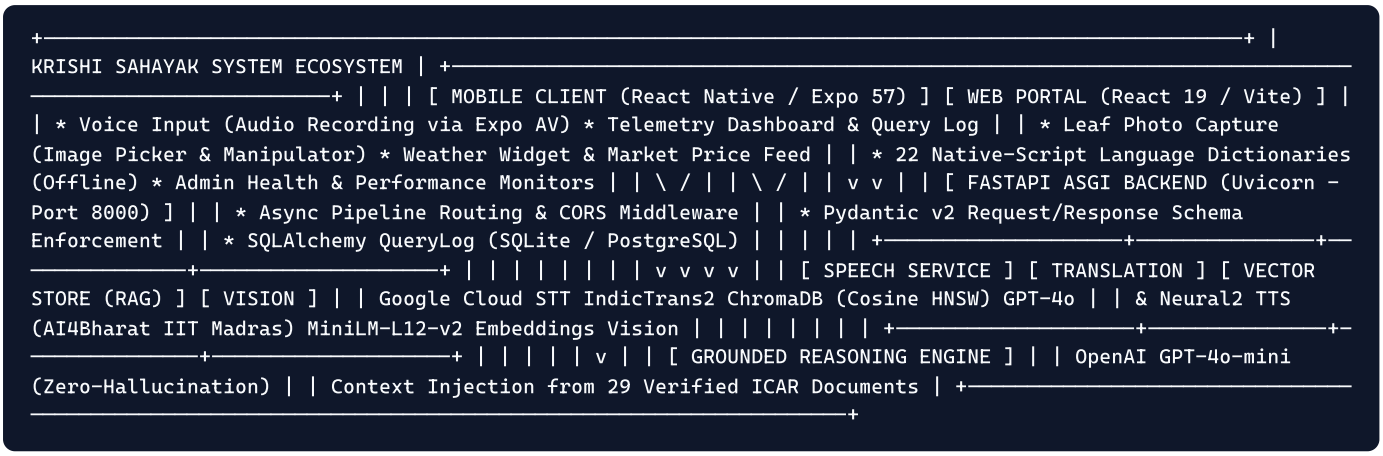
🗣️ Voice-First Oral Interface

Complete support for regional spoken input (Google Cloud STT / Whisper) and natural speech output (WaveNet / Neural2 TTS) enables illiterate farmers to receive spoken advice.

🌐 22 Official Languages in Native Script

Full offline client dictionaries and deep neural translation (IndicTrans2) cover all Eighth Schedule languages in their authentic scripts (Devanagari, Ol Chiki, Meitei, Gurmukhi, etc.).

2. System Architecture & Detailed Tech Stack



2.1 Tech Stack Breakdown

Layer / Component	Technology & Version	Architectural Role & Description
Mobile App Frontend	React Native 0.86.3 Expo SDK 57 (v57.0.26)	Cross-platform mobile client featuring file-based Expo Router (~57.0.24), native status bars, and accessibility features for rural farmers.
List Virtualization	`@shopify/flash-list` 2.0.2	High-performance 60 FPS list recycling utilizing dynamic automatic item measurement for instantaneous rendering of query history.
Media & Sensors	`expo-av` (~16.0.8) `expo-image-picker` (~57.0.20)	Handles audio capture (WEBM_OPUS / LINEAR16) and on-demand MP3 playback; manages camera reticle capture and image downsampling.
Web Portal	React 19.0.0 Vite 5.3.0, Tailwind CSS	Administrative and extension worker monitoring dashboard providing real-time query volume telemetry, weather insights, and advisory logs.
Backend Framework	FastAPI (Python 3.10+) Uvicorn ASGI Engine	High-concurrency asynchronous API service managing tri-modal endpoints (`/voice`, `/image`, `/text`) with Pydantic v2 validation.
Vector Database (RAG)	ChromaDB (PersistentClient) Cosine Similarity HNSW	Stores 384-dimensional vector embeddings across 29 curated ICAR packages of practice. Retains knowledge across container restarts.
Multilingual Embeddings	`paraphrase-multilingual-MiniLM-L12-v2`	500 MB multilingual sentence transformer embedding model enabling cross-lingual semantic matching between Indic queries and scientific manuals.
Grounded LLM	OpenAI GPT-4o-mini (Immutable Prompt)	Synthesizes practical, concise advisories (< 120 words) strictly constrained to retrieved context chunks with zero external training data hallucination.
Visual Diagnostics	OpenAI GPT-4o Vision	Objectively describes visible lesions, chlorosis, and insect infestations from crop photos, which are then used as queries for the RAG engine.
Neural Translation	IndicTrans2 (AI4Bharat) Google Cloud MT Fallback	Translates regional Indic language inputs into English as a search pivot, and translates verified advisories back into native scripts.
Speech Processing	Google Cloud STT Google Cloud Neural2 / WaveNet	Converts spoken regional dialect audio into text and converts final translated answers into high-fidelity streaming regional speech.
Relational Database	SQLite (`farmer.db`) SQLAlchemy 2.0 ORM	Maintains audit trails, user preferences, crop query logs, lat/long geolocation coordinates, and feedback ratings. Easily swappable to PostgreSQL.

3. End-to-End Data & Interaction Flow

Krishi Sahayak implements three multimodal ingestion pipelines (Voice, Image, Text) that converge into a unified RAG grounding and translation pipeline:

```
[ Farmer Voice Input ] [ Crop Leaf Photograph ] [ Farmer Text Input ] | | | v v v [ Google Cloud STT ] [ GPT-4o Vision ] [ Raw Regional Text ] Transcribes to native text Describes physical symptoms | | in English (No diagnosis) | v | | [ IndicTrans2 MT ] | | Translates to English pivot | | | | | +------------------------------+ | v [ English Semantic Search Query ] | v [ MiniLM-L12-v2 384-Dimensional Embedding ] | v [ ChromaDB Cosine Vector Similarity Search ] | v [ Top-K Verified ICAR Agronomic Chunks ] | v [ GPT-4o-mini Prompt Synthesis (Context Lock-In) ] * Strict zero-hallucination constraint * KVK referral fallback if context insufficient | v [ Verified Scientific Advisory in EN ] | v [ IndicTrans2: Reverse Translation to Native ] | v +--------------------+--------------------+ | | v (Screen Text) v (On-Demand Audio) [ Native Mobile UI ] [ Google WaveNet / Neural2 TTS ] Rendered via offline dicts Streamed as MP3 to farmer
```

3.1 Step-by-Step Query Execution Trace

1. **Input Acquisition:**
   - **Voice:** Farmer taps and holds the microphone button. Audio is recorded via `expo-av` (WEBM_OPUS for web, LINEAR16 for native) and sent as multipart data to `POST /api/v1/query/voice`.
   - **Image:** Farmer takes a photo of an infected leaf. `expo-image-manipulator` downsamples the image to 1024x1024 px JPEG (< 500 KB) and sends it to `POST /api/v1/query/image`.
   - **Text:** Farmer types a query in their native script or Latin transliteration and sends it to `POST /api/v1/query/text`.
2. **Speech Recognition & Vision Pre-Processing:**
   - Voice bytes are transcribed into native text using Google Cloud Speech-to-Text with regional language tags (e.g., `ml-IN`, `hi-IN`, `ta-IN`).
   - Image bytes are analyzed by **GPT-4o Vision** under a strict prompt: "*Describe only visible physical symptoms. Do NOT diagnose or suggest treatments.*" Output: "*Yellow halo surrounding dark necrotic lesions with concentric rings on lower leaves.*"
3. **Pivot Translation to Standard English:**
   - Native text queries are translated into English using **IndicTrans2** (IIT Madras). Using English as an intermediate semantic pivot prevents cross-lingual vector fragmentation and ensures alignment with official ICAR scientific research papers.
4. **Vector Embedding & ChromaDB Retrieval:**
   - The English query is encoded into a 384-dimensional dense vector using `paraphrase-multilingual-MiniLM-L12-v2`.
   - ChromaDB executes a cosine distance search across 29 curated ICAR knowledge manuals, extracting the top 3 verified text chunks with similarity scores > 0.40.
5. **Grounded Advisory Synthesis (Zero-Hallucination):**
   - The retrieved ICAR chunks and the query are passed to **GPT-4o-mini** with an immutable system prompt: "*Answer ONLY from the context provided below. If context is insufficient, return the standardized KVK referral message.*"
   - The LLM formats a practical response: Identification of cause, recommended organic/biological controls, verified chemical dosages (if necessary), and preventative cultural practices.
6. **Reverse Translation & Native Delivery:**
   - IndicTrans2 translates the English advisory back into the farmer's native regional language and native script.

- The mobile app renders the response inside `AdvisoryResponseSheet.tsx` with high-contrast text and action badges.

7. **Bandwidth-Efficient On-Demand Speech Synthesis:**
   - To preserve cellular data on 2G/3G connections, audio is not generated automatically. When the farmer taps the speaker icon, the app calls `GET /api/v1/query/{query_id}/audio`.
   - Google Cloud WaveNet / Neural2 TTS synthesizes natural spoken audio (MP3) and streams it directly to the device. Audio bytes are cached in memory via an LRU cache.

4. Codebase Structure & File Guide

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AI Based farmer advisory/ | ├── backend/ # FastAPI Asynchronous Microservice Backend | ├── app/ | | ├── api/ # HTTP API Route Controllers | | | ├── analytics.py # Telemetry, feedback scoring & query volume stats | | | ├── health.py # Liveness, readiness & external service probes | | | └── query.py # Tri-modal query ingestion & pipeline coordinator | | ├── models/ | | | └── query_log.py # SQLAlchemy QueryLog database entity | | ├── schemas/ | | | └── query.py # Pydantic v2 request/response validation contracts | | ├── services/ # Modular Business Logic & AI Engines | | | ├── cache.py # In-memory LRU audio and translation cache | | | ├── ingest.py # ICAR text chunking & ChromaDB vector ingestion | | | ├── llm.py # GPT-4o-mini grounded prompt inference & KVK referral | | | ├── market.py # Agmarknet / eNAM commodity wholesale price fetcher | | | ├── rag.py # ChromaDB PersistentClient & cosine similarity search | | | ├── speech.py # Google Cloud Speech-to-Text audio transcription | | | ├── translation.py # IndicTrans2 / Google MT translation engine | | | ├── tts.py # Google Cloud WaveNet/Neural2 speech synthesis | | | ├── vision.py # GPT-4o Vision physical symptom observer | | | └── weather.py # IMD / Open-Meteo agro-meteorology integration | | ├── config.py # Pydantic BaseSettings environment variable loader | | ├── database.py # SQLAlchemy database engine & scoped sessionmaker | | └── main.py # FastAPI application entrypoint & middleware setup | ├── knowledge_base/ # 29 Verified ICAR Agronomic Text Manuals | ├── farmer.db # SQLite persistent interaction store | └── requirements.txt # Backend Python dependencies | ├── frontend/ | ├── mobile/ # React Native / Expo SDK 57 Mobile App | | ├── app/ # Expo Router File-Based Navigation Tree | | | ├── (tabs)/ # Persistent Bottom Navigation Tabs | | | | ├── _layout.tsx # Tab bar shell & 3-step onboarding state machine | | | | ├── camera.tsx # Full-screen crop disease scanner & reticle UI | | | | ├── history.tsx # Optimized FlashList query interaction history | | | | └── index.tsx # Home dashboard: Voice, agro-weather, crop cards | | | └── _layout.tsx # Root layout, StatusBar & TextSize context provider | | ├── components/ # Modular Mobile UI Components | | | ├── AdvisoryResponseSheet.tsx # Modal sheet displaying verified recommendations | | | ├── FarmerLoginView.tsx # Mobile/OTP authentication & guest login | | | ├── LanguageSelectionView.tsx # 22-language native script grid selector | | | ├── LanguageSelectorSheet.tsx # Dynamic floating language switcher modal | | | ├── LocationPickerModal.tsx # State & District geolocation selector | | | └── OnboardingFlow.tsx # Initial farmer setup wizard | | ├── constants/ | | | └── india_locations.ts # Complete database of all 28 Indian States & Districts | | ├── locales/ # 23 Native-Script Localization Dictionaries (JSON) | | | ├── en.json, hi.json, ml.json, ta.json, te.json, bn.json, mr.json, gu.json ... | | | └── doi.json, brx.json, kok.json, ks.json, mai.json, mni.json, sat.json, ur.json | | ├── services/ # Mobile API & Utility Layer | | | ├── api.ts # Axios client with timeout & error interceptors | | | ├── haptics.ts # Safe cross-platform tactile feedback | | | ├── i18n.ts # Dynamic i18next translation resolution engine | | | ├── storage.ts # AsyncStorage abstraction for user preferences | | | └── weather.ts # Hyperlocal agro-weather forecast service | | ├── app.json # Expo configuration (permissions, plugins, icons) | | ├── metro.config.js # Metro bundler resolver configuration | | └── package.json # Mobile dependencies (Expo SDK 57, React 19) | | | └── web/ # React 19 / Vite Web Portal | ├── src/ | | ├── components/Dashboard.tsx # Comprehensive farmer & admin monitoring dashboard | | ├── services/api.ts # Web HTTP communication service | | ├── App.tsx # Top-level Web shell | | └── main.tsx # DOM mounting point | ├── package.json # Web dependencies (Vite, Tailwind, React 19) | └── vite.config.ts # Vite bundler build parameters | └── docker-compose.yml # Multi-container local deployment manifest
```

5. Localization Strategy: 22 Official Indian Languages

5.1 The Pan-India Linguistic Architecture

India's agricultural heartlands span 28 states with distinct scripts and dialects. Krishi Sahayak implements a dual-layer localization model:

- **Layer 1 (Client-Side Static UI):** 23 native-script JSON dictionaries stored locally in `frontend/mobile/locales/*.json`. Every dictionary contains 184 identical key-value pairs covering crop names, navigation, pest alerts, and weather cards. These render instantly with zero network requests.
- **Layer 2 (Server-Side Dynamic AI):** IndicTrans2 (IIT Madras) performs real-time neural translation between regional dialects and English for unstructured agronomic questions and generated answers.

5.2 Complete Language Coverage Matrix

Language	ISO Code	Native Script Name	Script Classification	Primary Agricultural Belts
English	`en`	English	Latin (Default First)	Pan-India / Administrative
Hindi	`hi`	हिन्दी	Devanagari	Uttar Pradesh, Madhya Pradesh, Bihar, Rajasthan
Malayalam	`ml`	മലയാളം	Malayalam	Kerala, Lakshadweep (Spices, Coconut, Rubber)
Bengali	`bn`	বাংলা	Eastern Nagari	West Bengal, Tripura, Assam (Paddy, Jute)
Telugu	`te`	తెలుగు	Telugu	Andhra Pradesh, Telangana (Chilli, Cotton, Paddy)
Tamil	`ta`	தமிழ்	Tamil	Tamil Nadu, Puducherry (Sugarcane, Banana)
Marathi	`mr`	मराठी	Devanagari	Maharashtra, Goa (Soybean, Cotton, Sugarcane)
Gujarati	`gu`	ગુજરાતી	Gujarati	Gujarat (Groundnut, Cotton, Spices)
Kannada	`kn`	ಕನ್ನಡ	Kannada	Karnataka (Coffee, Pulses, Millets)
Odia	`or`	ଓଡ଼ିଆ	Odia	Odisha (Paddy, Pulses, Horticulture)
Punjabi	`pa`	ਪੰਜਾਬੀ	Gurmukhi	Punjab, Haryana (Wheat, Basmati Rice)
Assamese	`as`	অসমীয়া	Eastern Nagari	Assam, Brahmaputra Valley (Tea, Rice)
Urdu	`ur`	اردو	Perso-Arabic (Nastaliq)	Jammu & Kashmir, Telangana, UP, Bihar
Sanskrit	`sa`	संस्कृतम्	Devanagari	Classical / Educational
Sindhi	`sd`	سنڌي / सिंधी	Perso-Arabic / Devanagari	Gujarat, Rajasthan, Maharashtra
Kashmiri	`ks`	کٲشُر	Perso-Arabic	Kashmir Valley (Apple, Saffron, Walnut)
Nepali	`ne`	नेपाली	Devanagari	Sikkim, North Bengal (Cardamom, Tea, Ginger)
Konkani	`kok`	कोंकणी	Devanagari	Goa, Coastal Karnataka, Maharashtra (Cashew, Areca)
Manipuri	`mni`	ꯃꯤꯇꯩꯂꯣꯟ	Meitei Mayek	Manipur, North-East (Organic Horticulture)
Bodo	`brx`	बड़ो	Devanagari	Bodoland Territorial Region, Assam
Dogri	`doi`	डोगरी	Devanagari	Jammu Region, Himachal Pradesh (Basmati, Maize)
Maithili	`mai`	मैथिली	Devanagari	Northern Bihar, Jharkhand (Makhana, Mango)
Santali	`sat`	ᱥᱟᱱᱛᱟᱲᱤ	Ol Chiki	Jharkhand, Odisha, West Bengal (Tribal Agriculture)

5.3 Fallback Mechanics

> **Resilience Guarantee:** The translation function `getTranslation(lang, key)` operates with a 3-tier fallback: `Selected Language Dict (e.g. sat.json) → Base English Dict (en.json) → Raw Key Literal`.

> This guarantees the mobile app never crashes, encounters null-pointer exceptions, or renders empty UI elements during network loss.

6. Safety, Verification & Zero-Hallucination Guardrails

Agricultural advisories carry real-world financial and ecological consequences. An incorrect chemical recommendation or dosage can cause plant toxicity or severe crop loss. Krishi Sahayak enforces five strict safety guardrails:

1. Immutable Hardcoded System Prompt

Located in `backend/app/services/llm.py`. The system prompt is hardcoded in Python and isolated from runtime input, preventing prompt injection attacks from adversarial queries.

2. Contextual Lock-In (Retrieval Constraint)

The LLM is explicitly instructed to disregard general pre-training data and answer **ONLY** using the provided context chunks. It is forbidden from speculating beyond retrieved ICAR manuals.

3. Standardized KVK Referral Fallback

If ChromaDB similarity search returns zero chunks matching the query threshold (> 0.40 cosine score), the engine returns: "*I don't have verified information about this in my knowledge base. Please contact your local Krishi Vigyan Kendra.*"

4. Two-Stage Vision Separation

GPT-4o Vision is strictly restricted to describing physical symptoms (lesions, chlorosis, pests). It never diagnoses diseases or prescribes chemicals directly, eliminating visual misclassification risks.

5. Extractive Non-LLM Fallback

If external LLM APIs experience outages or rate-limiting, an automated extractive parser synthesizes key advisory sentences directly from raw ICAR documents, providing 100% grounded continuity.

7. Current Implementation Status & Verification

Diagnostic Dimension	Current Status	Verification Results & Audit Outcome
Expo SDK Modernization	SDK 57 Stable	Upgraded from Expo SDK 51 to 57 (v57.0.26) on React 19.2.3 and React Native 0.86.3 with clean peer dependencies.
TypeScript Integrity	0 Errors	`npx tsc --noEmit` passed cleanly with zero type errors across all mobile components and services.
Metro Bundler Build	2,946 Modules Bundled	`npx expo export --platform web` completed in 1.6s. Zero missing dependencies or unresolvable imports.
Health Diagnostic Check	100% Local Pass	`npx expo-doctor` passed 21/21 checks. Excluded unmaintained legacy modules as recommended.
Offline Dictionary Integrity	23 Languages Active	All 22 Eighth Schedule Indian language files populated with 184 verified native keys.
Knowledge Base Coverage	29 ICAR Manuals	Covers major crops: Paddy, Wheat, Cotton, Sugarcane, Banana, Mango, Mustard, Groundnut, Millets, Soil Health, and IPM.